MILITARY SPECIFICATION

ELECTRONIC COMPONENT ASSEMBLY: 11738850 (POWER CONTROL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Electronic Component Assembly: 11738850 (Power Control) (see 0.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedure for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies. Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11738850	Electronic Component Assembly: 11738850 (Power Control)

PACKAGING DATA SHEET

11738850	Packaging of Electronic Component Assembly: 11738850 (Power Control)

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The electronic component assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing F11738850 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following Requirements, where applicable, of MIL-F-13926.

a. Order of precedence

b. Dimensions and tolerances

c. Part identification and marking

d. Electrical and electronic assemblies

e. Wrorkmanship

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F} and 90°F.

3.3.1 Loads, power, and signals. The assembly shall perform as specified herein when the loads, power, and signals of table I are applied as specified,

3.3.1.1 Battery power relay drive. With the power source (item 2.1) of table I applied to P1-11, the output voltage at P1-13 shall be 11.0 +3.5 Vdc. With P1-20 connected to P1-7, the output Voltage at P1-13 shall be +22.0 ± 3.0 Vdc. With the power source of table I connected to P1-11, P1-22 and P1-21, and with P1-20 connected to P1-7, the output voltage at P1-13 shall be +11.0 ± 3.5 Vdc.

3.3.1.2 Battery power switching. With the load (item, 1.2 of table I) connected to F1-17 and P1-18, the power source (item 2.1) of table I applied to P1-22 and P1-21, and the digital-type D signal of table I applied to P1-9, the output voltage at P1-18 shall be as shown in figure I. (This signal may be recurring at a frequency ≤ 100 Hertz),

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TABLE I

LOADS, POWER, AND SIGNALS

Item	Condition	Characteristics	Connections
1.	Loads		Connected between
1.1	Resistor	10 ohms (Ω) ±5 percent	P1-16 and P1-7 when specified
1.2	Resistor	2.5Ω ±5 percent	P1-17, P1-18 and P1-7 when specified
1.3	Resistor	200Ω ±5 percent	P1-13 and P1-7
1.4	Resistor	3 kilohms (kΩ) ±5 percent	P1-15 and P1-7
1.5	Resistor	3.6 kΩ ±5 percent	P1-10 and P1-7
2.	Power Source		
2.1	+24 Volts direct current (Vdc)	Tolerance: ±0.5 Vdc Ripple: ≤0.1 Vrms	Applied as specified herein with return to P1-7 (-)
3.	Signal Source		
3.1	Digital Type D	Logical one - open circuit Logical zero - -0.3 ±0.3 V Pulse width - 100 ±20μsec at 50 percent amplitude point Pulse repetition frequency - adjustable (see figure 1)	Applied as specified herein
3.2	+24 V	Pulse width - 100 ±20 μsec at 50 percent amplitude point Pulse repetition frequency - adjustable (see figure 2)	Applied as specified herein

3.3.1.3 Battery power. Q-switch voltage. With the load connedted to P1-16, and with the power source of table 1 applied to P1-22 and P1-21, the output voltage at P1-16 shall be +23.9 ± 1.5 Vdc.

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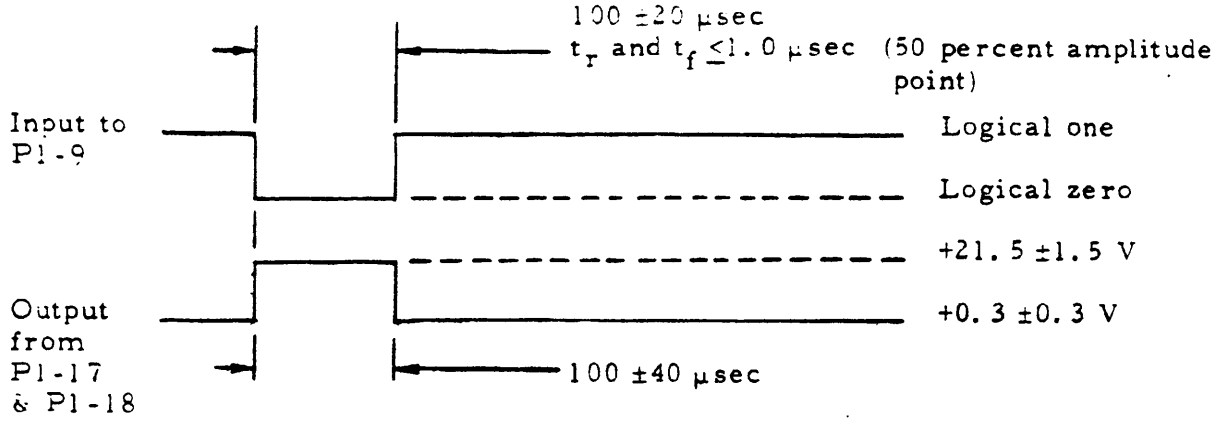


Figure 1. Battery Power Switching Waveforms

3.3.1.4 Batter power, voltage sense. With the power source of table I applied to P1-22, P1-21, and P1-11, the output voltage at P1-10, P1-8 and P1-15 shall be +12.0 ± 1.5 Vdc, +24.0 ± 1.5 Vdc respectively. With the power source of table I applied to P1-14, the output voltage at P1-15 shall be +11.5 ± 1.5 Vdc.

3.3.1.5 Tank power, primary power, and Q-switch voltage. With the loads (item 1.2 of table I) connected to P1-17 and P1-18, with the load (item 1.1 of table I) connected tp P1-16, and with the signal source (item 3.2) of table I applied to P1-24 and P1-25, and output at P1-17, P1-18 and P1-16 shall be as shown in figure 2.

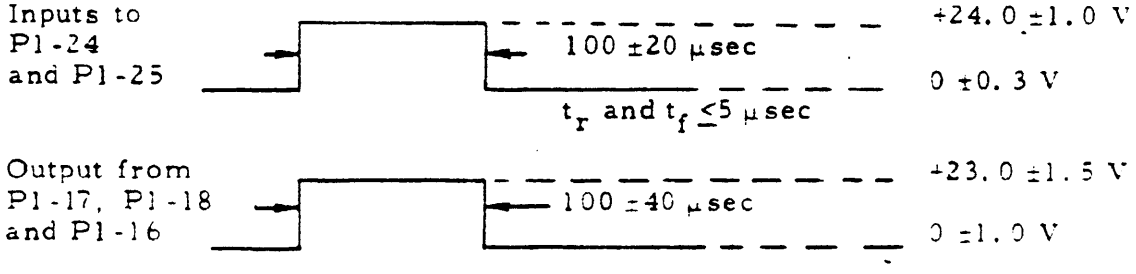


Figure 2. Tank Power Switching Waveforms

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3.3.2 Environmental

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80° F and -160°F.

3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1/32 inch double amplitude
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 to 5 Hertz in a 5-minute period
d. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specifies in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification.. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of th:s inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedure ?????? submission of product provisions of MIL-STD-105.

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4.3.2 Examination and test

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Suaaplementary quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPS, the app licable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Functional tests. The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

TABLE II
FUNCTIONAL TESTS

Classification	Characteristics	Requirement	Test Paragraph
101.	Battery power relay drive	3.3.1.1	4.6.2
102.	Battery power switching	3.3.1.2	4.6.3
103.	Battery power Q-switch voltage	3.3.1.3	4.6.4
104.	Battery power, voltage sense	3.3.1.4	4.6.5
105.	Tank power, primary power, and Q-switch voltage	3.3.1.5	4.6.6

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-SD-105.

4.4.1 General. One assemblies shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in table III.

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each year produced or from each month's production, whichever occured first. The control sample shall meet the requirments and test in table III.

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TABLE III
SPECIAL SAMPLING - GENERAL

Classification	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

TABLE IV
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Test	Requirement	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.7.1
304.	Operating temperature	3.3.2.2	4.6.7.1
305.	Vibration	3.3.2.3	4.6.7.2

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided bor by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table V shall be utilized in performing the tests specified herein.

TABLE V
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1	DC Power Supply	+24.0 ±0.5 Vdc Ripple: ≤0.1 Vrms Source impedance: ≤0.01 ohm
2	DC Voltmeter	0 to 50 Volts Accuracy ±1.0 percent

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4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPS forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4. 6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limitt and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MI MIL-F-13926.

4.6.1.1 Initial conditions. Apply the loads to the assembly as specified in table I.

4. 6.2 Battery power relay drive test. Apply the power source of table I (item 2.1 to F1-11. Measure the voltage at P1-13 tc determine compliance with 3.3.1.1. Connect P1-20 to P1-7. Measure the voltage at P1-13 to determine compliance with 3.3.1.1. Apply the power source of table I to P1-11, P1-22 and P1-21. Conect P1-20 to P1-7. Measure the output at P1-13 to determine compliance with 3.3.1.1.

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4.6.3 Batterv power switching test. Apply the load item 1.2 of of table I) to P1-1; and PI -18, apply the power source of table I item 2.11 to P1-22 and P1-21, and the digital type D signal (item 3.1 Of table I) to P1-9. Measure the output at P1-18 to determine compliance with 3.3.1.2.

4.6.4 Battery power, Q-switch voltage test. Connect the load item 1.1 of table I) to PI-16. Apply the power source of table I item 2.1 to P1-22 and P1-21. Measure the output at P1-16 to determine compliance with 3.3.1.3.

4.6.5 Battery power voltage sense test. Apply the power source of table I item 2.1) to P1-22, P1-21, and P1-11. Measure the outputs at P1-10, P1-8 and P1-15 to determine compliance with 3.3.1.4. Apply the power source to P1-14. Measure the output at P1-15 to determine compliance with 3.3.1.4.

4.6.6 Tank power, primary power, and Q-switch voltage test- Connect the loads (iterns 1.1 and 1.2 to P1-17, P1-18 and P1-16. Apply the power source of table I (item 2.1) to P1-24 and P1-25. Meaasure the output at P1-17, P1-18, and P1-16 to determine compliance with 3.3.1.5.

4.6.7 Enviromnental tests

4.6.7.1 Temperature. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 through 4.6.6 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.7.2 Vibration. Perform this test utilizing the apparatus speci - fied in Method 5.1.4.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposue, examine the assembly for structure failure, loose hardware, wire breakage, and abrason. Perform the tests of 4. 6.2 through 4.6.6 to determine compliance with 3.3.2.3.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
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